Mar. 24

- Defined last time
  - $G$ a group, $N \trianglelefteq G$ $\quad aN = Na \ \forall a \in G$
  - $G/N = \{aN : a \in G\}$ with group operation $(aN)\cdot(bN) = abN$

Ex. $\mathbb{Z}/m\mathbb{Z} = \{[a]_m : a \in \mathbb{Z}\}$

$\mathbb{Z}$ is Abelian, so every subgroup is normal

$[a]_m = a + m\mathbb{Z}$, $(a+m\mathbb{Z}) + (b+m\mathbb{Z}) = a+b+m\mathbb{Z}$

Ex $\langle r \rangle \triangleleft D_3$

$\quad \cong \qquad\quad \cong$

$A_3 \triangleleft S_3$

$\quad \cong$

$\{e, (123), (132)\}$

$S_3/A_3 = \{\overset{\bar{e}}{e\langle(123)\rangle}, \overset{\overline{(12)}}{(12)\langle(123)\rangle}\}$

$\langle(123)\rangle \quad (123)\langle(123)\rangle \quad (13)\langle(123)\rangle$

| $S_3/A_3$ | $\bar{e}$ | $\overline{(12)}$ |
|---|---|---|
| $\bar{e}$ | $\bar{e}$ | $\overline{(12)}$ |
| $\overline{(12)}$ | $\overline{(12)}$ | $\bar{e}$ |

$e\langle(123)\rangle \cdot e\langle(123)\rangle = (e\cdot e)\langle(123)\rangle = e\langle(123)\rangle = \bar{e}$

$(12)\langle(123)\rangle \cdot (12)\langle(123)\rangle = (12)(12)\langle(123)\rangle = e\langle(123)\rangle = \bar{e}$

Ex $\mathbb{R}/\mathbb{Z}$

$\frac{1}{2} + \mathbb{Z} + \frac{2}{3} + \mathbb{Z} = \frac{7}{6} + \mathbb{Z} = \frac{1}{6} + \mathbb{Z}$

real numbers "cycling back", reals mod integers

# Quotient Groups: Extended Example with $D_4$

Abstract Algebra

We can describe $D_4$, the dihedral group on four vertices, as

$$\langle s, r : s^2 = e,\ r^4 = e,\ sr = r^3 s\rangle.$$

$|D_4| = |H| \cdot [D_4 : H]$

$8 = 2 \cdot [D_4 : H]$

4 cosets

From this description we see that the elements are

$$\{e,\ r,\ r^2,\ r^3,\ s,\ sr = r^3 s,\ sr^2 = r^2 s,\ sr^3 = rs\}.$$

Consider the subgroup $H = \{e, r^2\}$. What are the left and right cosets of $H$ in $D_4$? Be sure you include all the different coset representations.

| left cosets | | right cosets |
|---|---|---|
| $eH = r^2H = H = \{e, r^2\}$ | $\longrightarrow\ =\ \longleftarrow$ | $He = Hr^2 = H = \{e, r^2\}$ |
| $rH = r^3H = \{r, r^3\}$ | $\longrightarrow\ =\ \longleftarrow$ | $Hr = Hr^3 = \{r, r^3\}$ |
| $sH = sr^2H = \{s, sr^2\}$ | $\longrightarrow\ =\ \longleftarrow$ | $Hs = Hr^2s = \{s, sr^2\}$ |
| $srH = sr^3H = \{sr, sr^3\}$ | $\longrightarrow\ =\ \longleftarrow$ | $Hsr = Hr^2sr = Hsr^3 = \{sr, sr^3\}$ |

Show that $H \triangleleft D_4$.

As shown above, the left and right cosets of $H$ in $D_4$ are equal for all elements of $D_4$

Complete the following Cayley table for $D_4/H$.

| $D_4/H$ | $eH$ | $sH$ | $rH$ | $srH$ |
|---|---|---|---|---|
| $eH$ | $e \cdot eH = eH$ | $sH$ | $rH$ | $srH$ |
| $sH$ | $sH$ | $ssH = eH$ | $sH \cdot rH = srH$ | $ssrH = rH$ |
| $rH$ | $rH$ | $rsH = srH$ | $r^2H = eH$ | $rsrH = sH$ |
| $srH$ | $srH$ | $srsH = rH$ | $sr^2H = sH$ | $srsrH = eH$ |

Complete the following Cayley table for $D_4$. Represent how the Cayley table on the previous page partitions this table.

| $D_4$ | $e$ | $r^2$ | $sr^2$ | $s$ | $r^3$ | $r$ | $sr$ | $sr^3$ |
|---|---|---|---|---|---|---|---|---|
| $e$ | $e$ | $r^2$ | $sr^2$ | $s$ | $r^3$ | $r$ | $sr$ | $sr^3$ |
| $r^2$ | $r^2$ | $e$ | $s$ | $sr^2$ | $r$ | $r^3$ | $sr^3$ | $sr$ |
| $sr^2$ | $sr^2$ | $s$ | $e$ | $r^2$ | $sr$ | $sr^3$ | $r^3$ | $r$ |
| $s$ | $s$ | $sr^2$ | $r^2$ | $e$ | $sr^3$ | $sr$ | $r$ | $r^3$ |
| $r^3$ | $r^3$ | $r$ | $sr^3$ | $sr$ | $r^2$ | $e$ | $sr^2$ | $s$ |
| $r$ | $r$ | $r^3$ | $sr$ | $sr^3$ | $e$ | $r^2$ | $s$ | $r^2$ |
| $sr$ | $sr$ | $sr^3$ | $r$ | $r^3$ | $s$ | $sr^2$ | $e$ | $r^2$ |
| $sr^3$ | $sr^3$ | $sr$ | $r^3$ | $r$ | $sr^2$ | $s$ | $r^2$ | $e$ |

What familiar direct product is $D_4/H$ isomorphic to?

$D_4/H = \{eH, sH, rH, srH\}$

$|D_4/H| = 4$ $\qquad$ $|\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}| = 4$

$$\begin{aligned}
|eH| = 1 &\xmapsto{\varphi} |(0,0)| = 1 \\
|sH| = 2 &\longmapsto |(1,0)| = 2 \\
|rH| = 2 &\longmapsto |(0,1)| = 2 \\
|srH| = 2 &\longmapsto |(1,1)| = 2
\end{aligned}$$

isomorphic to Klein 4-group, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$